Management Science

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Third-Party Consequences of Changes in Managerial Fiduciary Duties: The Case of Auditors' Going Concern Opinions

Liang Tan, Santhosh Ramalingegowda, Yong Yu

MANAGEMENT SCIENCE
Vol. 68, No. 2, February 2022, pp. 1556–1572
ISSN 0025-1909 (print), ISSN 1526-5501 (online)
http://pubsonline.informs.org/journal/mnsc

Third-Party Consequences of Changes in Managerial Fiduciary Duties: The Case of Auditors' Going Concern Opinions

Liang Tan,[a] Santhosh Ramalingegowda,[b] Yong Yu[c]

[a] Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061; [b] University of Georgia, Athens, Georgia 30602; [c] University of Texas at Austin, Austin, Texas 78712

Contact: liangtan@vt.edu, https://orcid.org/0000-0002-8421-3067 (LT); smr@uga.edu, https://orcid.org/0000-0002-1780-8810 (SR); yong.yu@mccombs.utexas.edu, https://orcid.org/0000-0003-4457-2820 (YY)

Received: August 14, 2017
Revised: May 5, 2019; April 14, 2020; August 28, 2020
Accepted: September 9, 2020
Published Online in Articles in Advance: March 29, 2021

https://doi.org/10.1287/mnsc.2020.3891

Abstract. This study examines the effect of managerial fiduciary duties on the likelihood of firms receiving going concern (GC) opinions from their auditors. We exploit an influential 1991 legal ruling that expanded fiduciary duties of corporate directors and officers in favor of creditors for near-insolvent Delaware firms. Our difference-in-differences test reveals an increase in GC opinions following the ruling for near-insolvent Delaware firms. Further tests indicate an increase in type I audit opinion errors and no change in audit risk after the ruling. Additional analysis shows that, after the ruling, near-insolvent Delaware firms are less likely to dismiss their auditors following the receipt of a GC report. Overall, our findings are consistent with managers and directors with increased fiduciary duties toward creditors exerting less pressure on auditors and allowing them to reveal more GC opinions. Our results highlight important third-party consequences of changes in managerial fiduciary duties.

History: Accepted by Shiva Rajagopal, accounting.

Keywords: Credit Lyonnais ruling • fiduciary duties • going concern opinions • auditors

1. Introduction

Fiduciary duties impose legal obligations and restrictions on the conduct of corporate directors and officers and constitute an important corporate-governance mechanism. Extant research on managerial fiduciary duties has focused largely on their effects on corporate policies such as investment, equity issuance, and financial-reporting decisions (e.g., Becker and Strömberg 2012 and Aier et al. 2014). In contrast, there is limited evidence of third-party consequences of managerial fiduciary duties. In this study, we attempt to bridge this gap by examining the effect of managerial fiduciary duties on the likelihood of firms receiving going concern (GC) opinions from their auditors.

As auditors' only direct form of communication, GC opinions convey new information that auditors discover about firms' equity and debt value, sending a warning signal to stakeholders (i.e., shareholders and debtholders), who can then take measures to protect their investments (e.g., Carson et al. 2013). GC opinions can be particularly beneficial to debtholders of distressed firms. GC opinions can trigger violations of debt covenants, resulting in a shift of control of the firm to debtholders (Menon and Williams 2010). As a result, managers and directors have incentives to pressure auditors to withhold issuing GC opinions (e.g., Carcello and Neal 2000 and Lennox 2000). Fiduciary duties legally bind directors and officers to act in the interest of the firm and stakeholders. If they derelict their fiduciary duties, stakeholders can sue them. Thus, fiduciary duties can be an effective instrument that incentivizes management to act in the best interest of the firm and stakeholders.[1] Consequently, an increase in managerial fiduciary duties toward debtholders should mitigate managerial incentives to pressure auditors to withhold GC opinions.

We exploit an influential 1991 legal ruling (the ruling, hereafter) that expanded fiduciary duties of corporate directors and officers in favor of creditors in near-insolvent Delaware firms. The ruling led near-insolvent Delaware firms to adopt more "creditor-friendly" policies, such as increased equity issues and investments, decreased operational risks (Becker and Strömberg 2012), and more conservative financial reporting (Aier et al. 2014, Bens et al. 2020). Thus, we expect that, after the ruling, managers of near-insolvent Delaware firms, considering creditors' interests, will likely exert less pressure on auditors and allow them to reveal more GC opinions. More specifically, we hypothesize that the likelihood of firms receiving a GC opinion increases after the ruling for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms.

To test our hypothesis, we adopt a difference-in-differences research methodology with firm fixed effects to examine changes in firms' likelihood of receiving GC opinions after the ruling. Using the default probability measure (Becker and Strömberg 2012) to identify near-insolvent firms, we find that, compared with near-insolvent non-Delaware firms, near-insolvent Delaware firms exhibit a significant increase in the likelihood of receiving GC opinions following the ruling. The effect of the ruling on GC opinions is also economically meaningful. Our results indicate that, after the ruling, the likelihood of receiving a GC opinion increases by 2.45 percentage points for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms. This increase corresponds to a nearly 35% increase from the unconditional likelihood of receiving a GC opinion in near-insolvent firms. Overall, these results are consistent with expanded fiduciary duties toward creditors increasing the likelihood of firms receiving GC opinions.

We next investigate whether the increase in managerial fiduciary duties toward creditors after the ruling results in higher audit risk for auditors. Prior research finds that auditors respond to higher audit risk by increasing audit effort (e.g., Ghosh and Tang 2015). Using audit-report lag as a proxy for auditor effort (Ghosh and Tang 2015), we find no change in audit effort after the ruling. This result suggests that the increase in GC opinions after the ruling is likely a result of a demand-side change in managers' incentive, rather than a supply-side change in audit effort in response to changes in audit risk.

We also examine the accuracy of the increased GC opinions after the ruling by investigating the type I and type II audit-opinion error rates (Francis 2011). Type I error occurs when the auditor issues a GC opinion, but the client firm does not subsequently fail. Type II error occurs when the auditor does not issue a GC opinion, but the client firm subsequently fails. We find that audit opinions contain more (less) type I (II) errors following the ruling. This finding is consistent with prior findings that auditors are conservative in general and err on the side of making type I errors when issuing audit opinions (e.g., Stanley et al. 2009 and Myers et al. 2014). More importantly, this finding suggests that the increased GC opinions after the ruling we document appear to be previously unobserved conservative GC opinions that auditors would have issued had there been less pressure from managers.

We subject our findings to a battery of robustness tests. First, we further examine causality. We find that the increase in GC opinions occurs only *after* the ruling, not *before*, suggesting that our primary finding is not the result of a reverse causality. Second, we examine the effect of the ruling in the long run. We find that the effect of the ruling gradually disappears in the long run, consistent with "leakage" of the ruling to other states. Third, we show that our results are robust to (1) using alternative insolvency cutoffs to identify near-insolvent firms; (2) using alternative measures of closeness to insolvency; (3) controlling for variations of the impacts of other economic events (e.g., the 1990–1991 recession) across geographic regions and industries; (4) controlling for media attention and reporting conservatism; and (5) matching GC with non-GC firms based on a propensity-score method to control for endogeneity.

Finally, to provide corroborating evidence that managers of near-insolvent Delaware firms exert less pressure on auditors issuing GC opinions after the ruling, we examine the likelihood of auditors being dismissed following the issuance of GC reports. One important way managers can influence auditor opinions is to dismiss the auditor who issues a GC report (e.g., Mutchler 1984 and Carcello and Neal 2003). If the ruling reduces managers' incentives to pressure auditors to withhold their GC opinions, we should observe that, after the ruling, auditors are less likely to be dismissed following the issuance of a GC report. Consistent with this expectation, we find that, relative to near-insolvent non-Delaware firms, the likelihood of auditors being dismissed following GC reports decreases after the ruling for near-insolvent Delaware firms.

Our paper contributes to the growing literature studying the economic consequences of managerial fiduciary duties. Although prior research has focused primarily on the effects of managerial fiduciary duties on corporate policies (e.g., Becker and Strömberg 2012 and Aier et al. 2014), we provide new evidence of third-party consequences of managerial fiduciary duties. We show that the effects of managerial fiduciary duties also extend to outcomes of other market participants, such as auditors' GC opinions.

Our study also adds to the auditing literature by examining the impact of an important governance mechanism—managerial fiduciary duties—on auditors' GC opinions. Although corporate-governance research in auditing has mainly focused on director independence, legally enforceable fiduciary duty is another powerful governance mechanism. Given the mixed evidence between director independence and firm performance (Bhagat and Black 2002), researchers argue that managerial incentives to avoid legal liability from breach of fiduciary duties may be a more effective governance mechanism (e.g., Gutiérrez 2003). Our findings that an exogenous increase in managerial fiduciary duties leads to more GC opinions and fewer dismissals of auditors who issue a GC report suggest that fiduciary duties play an important role in determining audit opinions.

The rest of the paper is organized as follows. Section 2 discusses our setting and hypothesis. Section 3 presents our research design, and Section 4 describes our sample and descriptive statistics.

Section 5 reports results, and Section 6 provides robustness tests and further analyses. Section 7 concludes.

2. Institutional Background and Hypothesis

2.1. Institutional Background

Under U.S. law, it is commonly understood that corporate directors and officers owe fiduciary duties to shareholders for a solvent firm, and their fiduciary duties expand to include creditors and other stakeholders only when the firm is insolvent. However, this position was changed in 1991 for all firms incorporated in Delaware by the Delaware court ruling in the landmark case of *Credit Lyonnais Bank Nederland v. Pathe Communication Corporation* (C.A. No. 12150, 1991 WL 277613).

In this case, Credit Lyonnais was a main creditor of MGM Corporation, and Pathe Communication was MGM's controlling shareholder. MGM filed for bankruptcy shortly after it underwent a leveraged buyout in November 1990. It then exited the bankruptcy, partly by obtaining a credit line from Credit Lyonnais, under an agreement that gave Credit Lyonnais the right to replace MGM's directors, including the CEO. Credit Lyonnais subsequently exercised this contractual right and replaced MGM's board. After the newly appointed directors rejected the request of Pathe Communication to sell certain assets, Pathe Communication sued Credit Lyonnais based on the claim that the board breached the fiduciary duty to MGM's shareholders by favoring interests of MGM's creditors. The Delaware Court ruled in December 1991 that the board had appropriately considered the potential differing interests between the corporation and its controlling shareholders, and "where a corporation is operating in the vicinity of insolvency, a board of directors is not merely the agent of the residual risk-bearers, but owes its duty to the corporate enterprise." In footnote 55 of the court's pronouncement, the judge further explained:

> The possibility of insolvency can do curious things to incentives, exposing creditors to risks of opportunistic behavior and creating complexities for directors . . . Such directors will recognize that in managing the business affairs of a solvent corporation in the vicinity of insolvency, circumstances may arise when the right (both the efficient and the fair) course to follow for the corporation may diverge from the choice that the stockholders (or the creditors, or the employees, or any single group interested in the corporation) would make if given the opportunity to act.

The ruling, especially footnote 55, attracted immediate, widespread media attention, triggered considerable amounts of comments and analyses, and became extensively cited by other cases, legal scholars, and lawyers (Becker and Strömberg 2012). Because the ruling did not provide a definitive meaning of "vicinity of insolvency," firms and lawyers faced uncertainty about exactly when directors and officers start to owe fiduciary duties to creditors and other stakeholders and to what extent. However, any uncertainty about the impact of this ruling on Delaware firms was generally viewed as "unidirectional" (Becker and Strömberg 2012), and the ruling was commonly accepted to have created a fiduciary duty of corporate directors and officers to parties other than shareholders for Delaware firms that approach insolvency (e.g., Hughes and McGee 1995 and Dionne 2007).[2]

In sum, the 1991 Delaware court ruling on the Credit Lyonnais case provides an arguably exogenous shock to managerial fiduciary duties for Delaware firms, which constitute about half of U.S. public corporations. In contrast, because Delaware courts have no prejudicial power for firms incorporated in other states, the 1991 Delaware court ruling did not affect managerial fiduciary duties for non-Delaware firms. Thus, this ruling creates a natural division of U.S. public firms into treatment (Delaware) and control (non-Delaware) groups and, thereby, facilitates a difference-in-differences design to identify the causal effects of managerial fiduciary duties. Further, we follow Becker and Strömberg (2012) to restrict our analyses to three years before and after the ruling to minimize the potential loss of power due to leakage (i.e., jurisdictions outside Delaware may learn from the Delaware ruling).

2.2. Hypothesis Development

Prior auditing literature (e.g., Francis and Wilson 1988, DeFond 1992, Carson et al. 2013, and DeFond and Zhang 2014) suggests that timely GCs can play an important role in protecting shareholders and debtholders. First, GCs send a warning signal about firms' equity and credit risk and liquidation value to shareholders and debtholders so that they can take quick measures to protect their investments. Holder-Webb and Wilkins (2000) find that GC opinions reduce investor surprises at bankruptcy announcements. Amin et al. (2014) find that GC opinions are positively associated with subsequent changes in cost of equity. Willenborg and McKeown (2000) find that initial public offerings with going-concern opinions experience less underpricing. Chen et al. (2016) show that upon receiving GCs, debtholders reduce the amount of lending, raise the interest rates of new loans, switch from financial covenants to more general covenants, and are more likely to require collateral. Second, GCs can trigger violations of debt covenants, resulting in a shift of control of the firm to debtholders (Menon and Williams 2010), which is beneficial to creditors of distressed firms.

Prior research also finds evidence consistent with managers pressuring auditors to withhold their GC opinions (Carcello and Neal 2000, Lennox 2000). Directors and officers can have significant influences on auditors' decisions to issue GC opinions. A GC opinion is the outcome of an often-contentious negotiation among managers, auditors, and audit committees. Directors and officers can pressure auditors to issue a clean opinion when a GC opinion is warranted. Prior research (Mutchler 1984, Lennox 2000, Carcello and Neal 2003) shows that auditors are more likely to be dismissed following the issuance of a GC opinion. In addition, directors and officers can also put pressure on audit fees, reduce the purchase of nonaudit services, and provide less cooperation during the audit process. Consistent with boards of directors influencing auditors' GC opinions, Carcello and Neal (2000) show that auditors are more likely to issue a GC opinion for firms with more independent directors on the audit committee, and Carcello and Neal (2003) find that auditors are less likely to be dismissed after issuing a GC opinion within firms whose directors on the audit committee have greater independence, more governance expertise, and lower shareholdings. The difficulty and ambiguity involved in the audit task of issuing a going-concern opinion might give auditors enough justification to succumb to management pressure and not issue a GC opinion when warranted (Carcello and Neal 2000).

Managerial fiduciary duties are likely to play a key role in determining the influences of directors and officers on GC opinions. Prior to the 1991 ruling, directors and officers of near-insolvent Delaware firms owed fiduciary duties to shareholders, but not to creditors or other stakeholders. Thus, directors and officers would more likely pressure auditors against issuing a GC opinion, especially when a GC opinion would violate debt covenants and result in a shift of control to creditors. After the 1991 ruling expanded managerial fiduciary duties to also take care of creditors for near-insolvent Delaware firms, we expect directors and officers to consider creditors' interests during financial distress, and, consequently, to be less likely to pressure auditors to withhold a GC opinion. We therefore expect near-insolvent Delaware firms to receive more GC opinions relative to near-insolvent non-Delaware firms following the ruling. Our hypothesis is as follows:

Hypothesis 1. *Near-insolvent Delaware firms are more likely to receive GC opinions than near-insolvent non-Delaware firms after the 1991 Credit Lyonnais ruling, ceteris paribus.*

However, there are also reasons to believe that the likelihood of near-insolvent Delaware firms receiving GC opinions may not change or may even decrease after the 1991 court ruling. First, prior to the ruling, auditors may not succumb to clients' pressure because of the reputational and litigation costs associated with the impaired independence (DeFond et al. 2002). Second, directors and officers have significant leeway to make decisions under the "business judgment rule." Thus, lawsuits against directors and officers for dereliction of fiduciary duties may not be effective, thereby mitigating their incentive to fulfil fiduciary duties.

3. Research Design

We conduct a difference-in-differences test to examine the effect of the ruling on the likelihood of firms receiving a going concern opinion (*GC*) for the first time. Specifically, we estimate the following regression model:

$$GC_{it} = \alpha_i + \lambda_t + \beta_1 DE_{it} \times POST_{it} + \beta_2 DE_{it} \times POST_{it} \times HIGHDP_{it} + \beta_3 HIGHDP_{it} + \gamma X_{it} + \varepsilon_{it}, \quad (1)$$

where we implement a linear probability model with firm fixed effects, and the subscripts i and t refer to firm and year respectively. α_i is a firm effect. λ_t is a year effect. The dependent variable, *GC*, is an indicator variable that equals one for firm-years that receive a first-time going concern audit opinion, and zero otherwise. *DE* is an indicator variable equal to one for firms incorporated in Delaware, and zero for firms incorporated outside Delaware. *POST* is an indicator variable equal to one for firm-years in the postruling window (i.e., 1992–1994) and zero for the preruling window (i.e., 1989–1991). *HIGHDP* is used to identify near-insolvent firms (Becker and Strömberg 2012, Aier et al. 2014, Bens et al. 2020).[3] It is an indicator variable that takes a value of one if a firm's *DP* is in the top quartile of the distribution, and zero otherwise, where *DP* is a firm's default probability derived from Merton's (1974) model, as described in Vassalou and Xing (2004). Our results are robust to using alternative cutoffs along the DP distribution and using alternative measures of closeness to default (see Section 6).

The key variable of interest in Equation (1) is the interaction term, $DE \times POST \times HIGHDP$, which captures changes in the likelihood of firms receiving a going-concern opinion from the preruling window to the postruling window for treatment (near-insolvent Delaware) firms relative to control (near-insolvent non-Delaware) firms. Our hypothesis predicts a significantly positive coefficient on $DE \times POST \times HIGHDP$ (i.e., $\beta_2 > 0$).

X is a vector of control variables. The *X* vector consists of a number of variables found in prior studies to determine GC opinions (e.g., Dopuch et al. 1987, Mutchler et al. 1997, Reynolds and Francis 2000, DeFond et al. 2002, and Chen et al. 2013). We include firm size (*SIZE*), which is the natural logarithm of total assets. Larger firms are expected to have more bargaining power to negotiate during financial distress and are less likely to go bankrupt (e.g., Reynolds and Francis 2000 and DeFond et al. 2002). We include leverage (*LEV*), which is total liabilities deflated by total assets, and change in leverage (*CLEV*) to proxy for the proximity to debt default. Prior studies find that debt defaults are associated with a higher likelihood of receiving GC opinions (e.g., Chen and Church 1992, Hopwood et al. 1994, and Mutchler et al. 1997). *LOSS* is an indicator variable that takes a value of one if a firm's income before extraordinary items is negative, and zero otherwise. Prior research finds that loss firms are more likely to fail (e.g., Reynolds and Francis 2000 and DeFond et al. 2002). *AGE* is the number of years that a firm has been publicly traded. Younger firms are expected to be more likely to receive a GC opinion (Dopuch et al. 1987).

We further control for a firm's stock return volatility (*VOLA*), which is the standard deviation of a firm's monthly stock returns over the year. We expect a positive association between return volatility and the likelihood of receiving a GC opinion. *RETURN* is a firm's stock return over the fiscal year. We expect a negative association between stock return and the likelihood of receiving a GC opinion. *LIQUIDITY* is the sum of a firm's cash and investment securities (long and short term) deflated by total assets. The liquidity measure proxies for a firm's ability to stave off bankruptcy when in financial distress. *OCF* is a firm's operating cash flows divided by total assets. Operating cash flow is expected to be negatively associated with the likelihood of bankruptcy. *NEWISSUE* indicates whether the firm has a new issuance of equity or debt in the subsequent year. Future financing has been found to be a mitigating factor to reduce the likelihood of bankruptcy (Mutchler et al. 1997). *BIGN* is an indicator variable that takes a value of one if a firm is audited by a Big N audit firm (Compustat item AU is between one and eight), and zero otherwise. *BETA* is a firm's beta estimated using a market model over the fiscal year. *REPORTLAG* is the number of days between the fiscal-year end date and the earnings-announcement date. The appendix provides detailed variable definitions. In all regression analyses, following Becker and Strömberg (2012), standard errors are clustered based on the interaction of year and state of incorporation.[4]

Our regression model (i.e., Equation (1)) examines GC opinions in both financially distressed and nondistressed firms. An advantage of this model is that it allows for the possibility of nondistressed firms becoming distressed at a later point of time. However, a potential concern is that the control variables we include in the model are from prior GC-opinion literature that focuses on only financially distressed firms. Although we are not aware of specific controls that affect GC opinions in nondistressed firms missing in our model, our difference-in-difference design and firm and year fixed effects mitigate concerns that any unknown correlated omitted variables bias our results. Nevertheless, we caution readers against interpreting the coefficients of control variables in our model for nondistressed firms.

4. Sample and Descriptive Statistics

Table 1 shows our sample-selection process. We start with all firm-years in the Compustat database over the period of 1989–1994, which contains three years before and three years after the ruling.[5] We next exclude financial, utility, and foreign firms. We require nonmissing data for state of incorporation. We drop firm-years that were already in default or bankruptcy filing when GCs were issued and firm-years with missing auditor code. Finally, we require each firm to have at least one observation in both preruling (1989–1991) and postruling (1992–1994) periods, and drop firm-years with missing values for control variables. These steps result in a final sample of 8,974 firm-years.

Auditors' GC opinions are hand-collected from the LexisNexis database. We first search the Securities and Exchange Commission (SEC) filings database in LexisNexis for the keyword of "going concern" over the period of 1988–1994.[6] This search yields 6,091 filings that mention the term of "going concern." We deliberately use this general keyword in the initial search in order to obtain a potential GC candidate sample as complete as possible. However, the cost of this approach is that the search also generates a lot of false positives, such as duplicate filings and discussions that contain the term of "going concern," but are

Table 1. Sample Selection

Sample selection	Firm-years
All Compustat firm-years from 1989 to 1994	62,965
Drop firm-years in financial and utility industries	(16,098)
Drop foreign firms and firm-years with missing Incorporation	(6,412)
Drop firm-years that were already in default or bankruptcy filing when GCs were issued and firm-years with missing auditor code	(5,007)
Drop firm-years without at least one observation in both pre and post periods	(5,727)
Drop firm-years with missing control variables	(20,747)
Number of firm-years in the final sample	8,974

Note. This table describes sample selection process to derive our sample.

unrelated to an audit opinion. To remove those filings unrelated to GC audit reports, we manually go through each filing to identify whether the filing indeed contains a GC audit report. As discussed in the previous paragraph, we also make sure that the GC opinions identified exclude those issued to a client firm that was already in default or had filed bankruptcy. Out of our final sample of 8,974 firm-years, 217 firm-years receive a first-time GC opinion. A total of 158 of them are near-insolvent (i.e., *HIGHDP* = 1) firm-years. The percentage of GC opinions for the near-insolvent sample is 7.04%. The statistic is consistent with prior GC studies of distressed firms (e.g., 8.29% in DeFond et al. 2002; 6.50% in Chen et al. 2013). As a comparison, for the 6,731 solvent firm-years, only 59 firm-years (0.88%) receive a first-time GC opinion.

Table 2 presents descriptive statistics for the variables used in Equation (1) for the final sample. All continuous variables are winsorized at the 1% and 99% levels to mitigate the influence of outliers. The average probability of receiving a first-time GC opinion (*GC*) in the full sample is 2.4%. Mean of *HIGHDP* is 25%, because we define near-insolvent as the top quartile of the default probability distribution. Around 31% of the firm-years in our sample have a loss (*LOSS*), and 62% of the firm-years have a new issuance of equity or debt in the subsequent year (*NEWISSUE*). The average firm age in the sample (*AGE*) is 16 years. The vast majority of our sample firms (90.3%) were audited by a *BIGN* auditor. All other variables have values that are generally consistent with prior studies. Table 3 presents Spearman correlation among the variables in the near-insolvent sample.

Table 2. Descriptive Statistics

Variable	*N*	Mean	SD	25th	Median	75th
GC	8,974	0.024	0.154	0.000	0.000	0.000
HIGHDP	8,974	0.250	0.433	0.000	0.000	0.000
SIZE	8,974	4.746	1.950	3.330	4.565	5.974
LEV	8,974	0.545	0.267	0.368	0.532	0.682
LOSS	8,974	0.309	0.462	0.000	0.000	1.000
VOLA	8,974	0.142	0.081	0.085	0.123	0.174
NEWISSUE	8,974	0.619	0.486	0.000	1.000	1.000
AGE	8,974	16.113	12.076	6.000	11.000	25.000
RETURN	8,974	0.160	0.678	−0.259	0.039	0.380
CLEV	8,974	0.008	0.137	−0.041	−0.002	0.049
LIQUIDITY	8,974	0.122	0.157	0.018	0.060	0.163
OCF	8,974	0.046	0.157	0.008	0.069	0.122
BIGN	8,974	0.903	0.296	1.000	1.000	1.000
BETA	8,974	1.034	1.597	0.213	0.979	1.783
REPORTLAG	8,974	57.926	35.529	36.000	51.000	74.000

Notes. This table presents descriptive statistics for the final sample. All variables are defined in the appendix.

5. Results

5.1. Univariate Analysis

Table 4 presents univariate analysis. We compare variables in the postruling period with those in the preruling period for treatment (near-insolvent Delaware) group and control (near-insolvent non-Delaware) group separately. We then calculate difference-in-differences for each variable by comparing the change in the treatment group with that in the control group. The univariate difference-in-differences (diff-in-diff) test shows a significant increase in firms receiving GC opinions following the ruling. For the treatment group, *GC* increases significantly following the ruling (p-value = 0.013), while we do not observe any significant changes in *GC* following the ruling for the control group (p-value = 0.443). This results in a significant positive diff-in-diff statistic (diff-in-diff = 0.048, p-value = 0.027), consistent with Hypothesis 1 in that relative to near-insolvent non-Delaware firms, near-insolvent Delaware firms receive more GC opinions following a shift in fiduciary duties to creditors. Most of the control variables do not show significant difference-in-differences, except for *BIGN* and *BETA*. These differences will be controlled for in the regression analysis discussed in Section 5.2.

5.2. The Effect of the Ruling on the Likelihood of Firms Receiving Auditors' GC Opinions

Table 5 presents results of estimating Equation (1) to examine the effect of the ruling on the likelihood of firms receiving auditors' first-time GC reports.[7] We find a significantly positive coefficient on our variable of interest – the interaction term $DE \times POST \times HIGHDP$ (coefficient = 0.056, t-stat = 3.14). This result indicates that, relative to near-insolvent non-Delaware firms, near-insolvent Delaware firms are more likely to receive a GC audit opinion after the ruling, supporting our hypothesis (Hypothesis 1). We also note that the interaction term $DE \times POST$ is insignificant, and the coefficient is close to zero. These findings indicate that solvent Delaware firms show no significant changes in the likelihood of receiving GC opinions relative to solvent non-Delaware firms following the ruling, providing a nice placebo test result further supporting our hypothesis.

To gauge economic significance of the ruling effect, we rerun our analysis using a probit model and then calculate average marginal effects.[8] The average marginal effect of our variable of interest is 0.0245. Thus, relative to near-insolvent non-Delaware firms, near-insolvent Delaware firms are 2.45% more likely to receive an auditor's GC opinion after the ruling. This corresponds to a nearly 35% increase from the unconditional likelihood of receiving a GC opinion for near-insolvent firms (i.e., 2.45%/7.04%). This result

Table 3. Correlations

	Variable	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]
[1]	*GC*	1.000													
[2]	*SIZE*	0.007	1.000												
[3]	*LEV*	0.193	0.251	1.000											
[4]	*LOSS*	0.188	−0.076	0.157	1.000										
[5]	*VOLA*	0.082	−0.336	0.086	0.144	1.000									
[6]	*NEWISSUE*	−0.018	0.156	0.190	−0.005	−0.011	1.000								
[7]	*AGE*	0.040	0.213	0.133	−0.063	−0.156	−0.013	1.000							
[8]	*RETURN*	−0.129	0.032	0.052	−0.257	0.104	0.050	0.137	1.000						
[9]	*CLEV*	0.216	0.011	0.346	0.445	0.033	0.048	−0.050	−0.273	1.000					
[10]	*LIQUIDITY*	−0.039	−0.112	−0.240	0.045	0.072	−0.131	0.004	−0.059	−0.044	1.000				
[11]	*OCF*	−0.107	0.097	−0.080	−0.328	−0.126	−0.044	0.049	0.181	−0.282	−0.043	1.000			
[12]	*BIGN*	0.020	0.311	0.055	−0.014	−0.099	0.069	0.040	0.038	−0.013	0.005	0.064	1.000		
[13]	*BETA*	−0.041	0.150	−0.007	−0.070	0.093	0.011	−0.000	−0.040	0.003	0.033	0.040	0.062	1.000	
[14]	*REPORTLAG*	0.194	−0.380	0.157	0.283	0.230	0.020	−0.099	−0.084	0.166	−0.028	−0.177	−0.227	−0.102	1.000

Notes. This table presents descriptive statistics for the Spearman correlation table for the sample that is near-insolvent. All variables are defined in the appendix.

suggests that the effect of the ruling on the likelihood of firms receiving GC opinions is economically meaningful.

Results on the control variables are generally consistent with our predictions and prior research. We find that firms are more likely to receive GC opinions when they have losses (*LOSS*), higher leverage (*LEV*), and an increase in leverage (*CLEV*); have Big N auditors (*BIGN*); and have longer reporting lags (*REPRTLAG*). On the other hand, firms are less likely to receive GC opinions when they have larger amounts of cash and investment securities (*LIQUIDITY*) and higher operating cash flows (*OCF*).

Overall, the results from Table 5 provide consistent evidence in support of our Hypothesis 1. These results show that the *Credit Lyonnais* ruling leads to a significant increase in the likelihood of firms receiving GC reports for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms.

5.3. Audit Risk and Changes in the Accuracy of GC Reports After the Ruling

In this subsection, we first investigate whether the increase in managerial fiduciary duties toward creditors results in higher audit risk for the auditors. Prior research finds that auditors respond to higher audit risk by increasing effort (e.g., Ghosh and Tang 2015). Following this research, we use audit-report lag as a proxy for auditor effort. We estimate Equation (1) by

Table 4. Univariate Analysis

	Delaware			Non-Delaware				
Variable	Pre	Post	*p*-value	Pre	Post	*p*-value	Diff-in-diff	*p*-value
Firm-years	677	609		506	451			
GC	0.053	0.089	0.013	0.077	0.065	0.443	0.048	0.027
SIZE	4.524	4.150	0.000	3.929	3.408	0.000	0.146	0.284
LEV	0.723	0.730	0.686	0.678	0.696	0.421	−0.010	0.725
LOSS	0.598	0.616	0.521	0.585	0.652	0.034	−0.049	0.236
VOLA	0.186	0.194	0.134	0.181	0.197	0.015	−0.008	0.335
NEWISSUE	0.662	0.632	0.268	0.595	0.568	0.394	−0.002	0.955
AGE	13.793	13.496	0.622	15.087	13.710	0.030	1.080	0.224
RETURN	−0.280	−0.138	0.000	−0.294	−0.174	0.001	0.022	0.662
CLEV	0.034	0.031	0.807	0.035	0.051	0.183	−0.018	0.250
LIQUIDITY	0.077	0.085	0.248	0.074	0.089	0.032	−0.007	0.471
OCF	0.032	0.003	0.002	0.028	−0.002	0.005	0.000	0.974
BIGN	0.900	0.888	0.514	0.862	0.800	0.011	0.050	0.083
BETA	0.938	0.931	0.950	0.877	0.528	0.005	0.342	0.043
REPORTLAG	74.854	76.828	0.409	79.476	79.528	0.987	1.922	0.618

Notes. This table presents univariate comparison between Delaware firms (treatment group) and non-Delaware firms (control group) pre and post the *Credit Lyonnais* ruling. The sample is restricted to firms that are close to insolvency (i.e., *HIGHDP* = 1). All variables are defined in the appendix.

Table 5. The Effect of the Ruling on the Likelihood of Firms Receiving Auditors' GC Opinions

	GC	
Variable	Coefficient	*t*-stat
DE × *POST*	−0.000	(−0.03)
DE* × *POST* × *HIGHDP	**0.056*** **	**(3.14)**
SIZE	0.005	(0.95)
LEV	0.073***	(2.76)
LOSS	0.025***	(4.56)
VOLA	−0.035	(−0.72)
NEWISSUE	−0.005	(−1.22)
AGE	0.000	(0.40)
RETURN	−0.004	(−1.23)
CLEV	0.066***	(3.75)
HIGHDP	0.000	(0.04)
LIQUIDITY	−0.034**	(−2.35)
OCF	−0.065**	(−2.19)
BIGN	0.047***	(3.14)
BETA	−0.001	(−0.53)
REPORTLAG	0.000***	(2.64)
Firm effects	Yes	
Year effects	Yes	
Observations	8,974	
Adjusted R^2	0.04	

Notes. This table presents the effect of the *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

replacing *GC* with *REPORTLAG* and controlling for variables that determine *REPORTLAG*. Table 6 reports the results. The coefficient on our variable of interest $DE \times POST \times HIGHDP$ is insignificant (coefficient = −0.230, *t*-stat = −0.14), indicating no changes in audit effort for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms following the ruling. This finding suggests that our earlier results of an increase in GC opinions after the ruling is unlikely to be an artifact of any supply-side change in audit effort in response to a change in audit risk.

We next examine the accuracy of increased GC opinions after the ruling by investigating the type I and type II audit opinion error rates (Francis 2011). Type I error occurs when the auditor issues a GC opinion, but the client firm does not subsequently fail. The type I error rate is the $PROB\ (GC = 1 \mid Failure = 0)$. Therefore, to examine changes in type I error rate following the ruling, we estimate Equation (1) conditional on $Failure = 0$. A significantly positive (negative) coefficient on $DE \times POST \times HIGHDP$ will indicate an increase (decrease) in type I error rate for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms after the ruling, while an insignificant coefficient will indicate no change.

Type II audit opinion error occurs when the auditor does not issue a GC opinion, but the client firm subsequently fails. The probability of making a type II error in a test is $1 - PROB\ (GC = 1 \mid Failure = 1)$, where $PROB\ (GC = 1 \mid Failure = 1)$ is power of the test. Therefore, estimating Equation (1) conditional on $Failure = 1$ will give us changes in the power of the test for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms after the ruling. Because one minus the power of the test is type II error rate, the higher the power of the test, the lower the type II error rate. Therefore, a significantly positive (negative) coefficient on $DE \times POST \times HIGHDP$ will indicate a decrease (increase) in type II error rate for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms after the ruling, while an insignificant coefficient will indicate no change.

Table 6. Changes in Audit Effort After the Ruling

Variable	*REPORTLAG*	*t*-stat
DE × *POST*	−0.874	(−1.06)
DE* × *POST* × *HIGHDP	**−0.230**	**(−0.14)**
SIZE	−4.828***	(−4.13)
ROA	−26.114***	(−6.24)
LEV	12.444***	(3.75)
OPINION	−2.891***	(−4.37)
AUCHANGE	1.166	(1.20)
BUSY	3.428**	(2.03)
DA	1.451	(0.40)
HINDEX	5.516	(1.39)
FSALE	5.445	(1.01)
Firm effects	Yes	
Year effects	Yes	
Observations	8,829	
Adjusted R^2	0.06	

Notes. Table 7 and this table present results examining changes in audit effort and the accuracy of GC reports after the ruling. This table presents results of the audit effort test. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

We consider a firm to have subsequently failed if the firm either filed for bankruptcy or was delisted from the stock exchange (delisting code is 4 or 5) within one year of an audit. Table 7 reports the results. For the type I error test, the coefficient on our variable of interest, $DE \times POST \times HIGHDP$, is positive and statistically significant (coefficient = 0.036, *t*-stat = 2.88), consistent with a higher type I audit opinion error rate for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms after the ruling. For the type II audit error test, the coefficient on $DE \times POST \times HIGHDP$ is also positive and statistically significant (coefficient = 0.844, *t*-stat = 3.13), consistent with a lower type II audit opinion error rate for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms after the ruling. In sum, these findings are consistent with prior findings

Table 7. Changes in the Accuracy of GC Reports After the Ruling: Type I and Type II Error Rate Test

	Type I error test		Type II error test	
	GC | *FAILURE* = 0		*GC* | *FAILURE* = 1	
Variable	Coefficient	*t*-stat	Coefficient	*t*-stat
DE × *POST*	−0.001	(−0.14)	−0.548	(−1.37)
DE* × *POST* × *HIGHDP	**0.036*****	**(2.88)**	**0.844*****	**(3.13)**
SIZE	0.008	(1.60)	0.120	(0.94)
LEV	0.048**	(2.12)	−0.506***	(−3.94)
LOSS	0.023***	(4.46)	0.234**	(2.38)
VOLA	−0.042	(−0.96)	1.302***	(3.60)
NEWISSUE	0.001	(0.28)	−0.016	(−0.27)
AGE	−0.000	(−0.20)	−0.040	(−0.83)
RETURN	−0.003	(−1.25)	−0.137***	(−2.69)
CLEV	0.055***	(2.75)	0.667***	(5.54)
HIGHDP	0.003	(0.48)	−0.695***	(−3.80)
LIQUIDITY	−0.015	(−0.90)	−1.205	(−1.18)
OCF	−0.057*	(−1.93)	−0.781*	(−1.99)
BIGN	0.039**	(2.36)	0.610***	(3.25)
BETA	−0.001	(−0.77)	−0.005	(−0.18)
REPORTLAG	0.000**	(2.24)	0.002*	(1.74)
Firm effects	Yes		Yes	
Year effects	Yes		Yes	
Observations	8,758		216	
Adjusted R^2	0.04		0.41	

Notes. Table 6 and this table present results examining changes in audit effort and the accuracy of GC reports after the ruling. This table presents results of the type I and type II error rate test. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

that auditors are conservative in general and tend to err on the side of making type I errors when issuing audit opinions (e.g., Stanley et al. 2009 and Myers et al. 2014). The increased GC opinions after the ruling appear to be previously unobserved conservative opinions that auditors would have issued had there been less pressure from their client management.

6. Robustness Tests and Further Analyses

6.1. A Dynamic Model—Test of Parallel Trend and the Effect in the Long Run

We estimate a "dynamic" model as in Bertrand and Mullainathan (2003) to examine the parallel trend assumption in the difference-in-differences methodology and explore the effect of the ruling in the long run. The parallel trend assumption implies that in the absence of the treatment effect (i.e., the ruling), GC reports of near-insolvent Delaware and non-Delaware firms would have changed at the same rate over time. Further, it is not clear ex ante whether the effect of the ruling will persist in the long run. To the extent that other states start to recognize the ruling as important legal precedent or incorporate legislative ideas from Delaware (Becker and Strömberg 2012), the effect of the ruling may disappear over time (i.e., the leakage of the ruling to other states).

Table 8 reports our findings. In column (1), we closely follow Bertrand and Mullainathan (2003) and replace the *POST* dummy with four year-specific indicators: $BEFORE^{-1}$, $BEFORE^{0}$, $AFTER^{1}$, and $AFTER^{2+}$, where $BEFORE^{-1}$ is an indicator variable equal to one if the court ruling will pass one year from now. $BEFORE^{0}$ is an indicator variable equal to one if the court ruling passed this year, and $AFTER^{1}$ and $AFTER^{2+}$ are indicator variables equal to one if the court ruling passed one year ago and two or more years ago, respectively. We find that the increase in GC opinions occurs only *after* the ruling ($DE \times AFTER^{1} \times HIGHDP$ and $DE \times AFTER^{2+} \times HIGHDP$ are significant), not *before* the ruling ($DE \times BEFORE^{-1} \times HIGHDP$ and $DE \times BEFORE^{0} \times HIGHDP$ are insignificant), supporting the parallel trend assumption. Moreover, this result also mitigates the concern of reverse causality.

Column (2) further breaks down $DE \times AFTER2^{+} \times HIGHDP$ variable into $DE \times AFTER^{2} \times HIGHDP$ and $DE \times AFTER^{3} \times HIGHDP$. We see that $DE \times AFTER^{3} \times HIGHDP$ (i.e., the effect in 1994) starts to lose significance (t-stat = 1.65; p-value is just below the traditional two-tailed 10% threshold). Column (3) extends our testing window to 1995, and we find that the effect in 1995 ($DE \times AFTER^{4} \times HIGHDP$) is insignificant. Taken together, the results from columns (2) and (3) indicate that the effect of the ruling disappears gradually over time, consistent with leakage of the ruling to other states over time.

6.2. Alternative Cutoffs to Identify Firms Close to Insolvency

So far, we follow prior literature (e.g., Aier et al., 2014) and use the top quartile of the default probability distribution as the cutoff to identify firms close to insolvency. However, it is not clear from the ruling itself as to what constitutes the "vicinity of insolvency." We therefore assess the robustness of our results by using alternative cutoff points along the default probability distribution. Specifically, we con sider two alternatives: top tertile (*HIGHDP33*) and top quintile (*HIGHDP20*), which represent one wider and one narrower cutoff along the default probability distribution, respectively. Table 9 reports the results of estimating Equations (1) using the two alternative cutoffs for near-insolvent firms. We continue to find an increase in the likelihood of firms receiving GC opinions for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms, indicating that our results are insensitive to the cutoffs we use to identify firms in the vicinity of insolvency.

Table 8. A Dynamic Model—Test of Parallel Trend and the Effect in the Long Run

	(1)		(2)		(3)	
Variable	GC	*t*-stat	GC	*t*-stat	GC	*t*-stat
$DE \times BEFORE^{-1}$	−0.005	(−0.63)	−0.005	(−0.63)	−0.005	(−0.63)
$DE \times BEFORE^{0}$	−0.002	(−0.21)	−0.002	(−0.21)	0.000	(0.02)
$DE \times AFTER^{1}$	−0.003	(−0.43)	−0.004	(−0.45)	−0.002	(−0.25)
$DE \times AFTER^{2+}$	−0.001	(−0.15)				
$DE \times AFTER^{2}$			−0.009	(−1.07)	−0.006	(−0.81)
$DE \times AFTER^{3}$			0.007	(0.99)	0.011	(1.38)
$DE \times AFTER^{4}$					−0.012	(−1.44)
$DE \times BEFORE^{-1} \times HIGHDP$	0.003	(0.21)	0.003	(0.21)	−0.001	(−0.09)
$DE \times BEFORE^{0} \times HIGHDP$	0.017	(1.40)	0.017	(1.42)	0.009	(0.70)
$\boldsymbol{DE \times AFTER^{1} \times HIGHDP}$	**0.081*****	**(7.94)**	**0.082*****	**(8.01)**	**0.075*****	**(7.85)**
$\boldsymbol{DE \times AFTER^{2+} \times HIGHDP}$	**0.044****	**(2.47)**				
$\boldsymbol{DE \times AFTER^{2} \times HIGHDP}$			**0.057*****	**(4.99)**	**0.039*****	**(2.95)**
$\boldsymbol{DE \times AFTER^{3} \times HIGHDP}$			**0.031**	**(1.65)**	**0.012**	**(0.63)**
$\boldsymbol{DE \times AFTER^{4} \times HIGHDP}$					**0.017**	**(1.12)**
SIZE	0.004	(0.71)	0.003	(0.64)	0.001	(0.19)
LEV	0.073***	(2.78)	0.073***	(2.76)	0.094***	(3.85)
LOSS	0.025***	(4.54)	0.025***	(4.54)	0.022***	(5.03)
VOLA	−0.044	(−0.87)	−0.044	(−0.86)	−0.013	(−0.27)
NEWISSUE	−0.005	(−1.20)	−0.005	(−1.19)	−0.005	(−1.13)
AGE	0.001	(0.55)	0.000	(0.10)	0.016***	(3.12)
RETURN	−0.004	(−1.22)	−0.004	(−1.17)	−0.006*	(−1.94)
CLEV	0.065***	(3.66)	0.065***	(3.70)	0.069***	(4.39)
HIGHDP	−0.002	(−0.25)	−0.002	(−0.24)	0.002	(0.30)
LIQUIDITY	−0.033**	(−2.30)	−0.033**	(−2.29)	−0.014	(−0.76)
OCF	−0.064**	(−2.15)	−0.064**	(−2.14)	−0.059**	(−2.28)
BIGN	0.047***	(3.13)	0.047***	(3.15)	0.019	(1.25)
BETA	−0.001	(−0.41)	−0.001	(−0.38)	0.001	(0.83)
REPORTLAG	0.000***	(2.62)	0.000***	(2.63)	0.000***	(2.62)
Firm effects	Yes		Yes		Yes	
Year effects	Yes		Yes		Yes	
Observations	8,974		8,974		10,893	
Adjusted R^2	0.04		0.04		0.05	

Notes. This table presents parallel trend and long-run effect analyses of the effect of the *Credit Lyonnais* ruling on GC opinions. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.
*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

6.3. Alternative Measures for Insolvency

In addition to the default probability (*DP*) measure, we also examine *ZSCORE* and *LEV* as alternative proxies for closeness to insolvency. Because a lower (higher) *ZSCORE* (*LEV*) indicates a higher default risk, we classify firms as near-insolvent firms if their *ZSCORE* (*LEV*) is in the bottom (top) quartile of the distribution, and firms as solvent firms if their *ZSCORE* (*LEV*) is above (below) the bottom (top) quartile of the distribution. As shown in Table 10, for both measures, we continue to find a significant increase in the likelihood of receiving GC opinions after the ruling for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms.

6.4. Controlling for Variations Across Geographic Regions and Industries

Our sample period includes economic events such as the 1990–1991 recession. To the extent that the impact of these events varies across geographic regions and industries, they may affect our results. For example, if the 1990–1991 recession affected the U.S. northeast more than other regions, it could affect Delaware firms more than non-Delaware firms and, thus, possibly drive some of the differences we have observed between the two sets of firms. To address this issue, we follow Becker and Strömberg (2012) and include *Year* × *State* fixed effects. Note that a state here refers to a firm's headquarter state, not state of incorporation, and the assumption is that a firm's headquarter state captures its exposure to the regional business cycles. These additional fixed effects account for the heterogeneity in each year and the headquarter state of each firm, and, thereby essentially compare Delaware and non-Delaware firms headquartered in the same state each year. Same as Becker and Strömberg (2012), we also include *Year* × *Industry* fixed effects to ensure that our results are not affected by the

Table 9. Alternative Insolvency Cutoffs

	(1)		(2)	
Variable	GC	*t*-stat	GC	*t*-stat
DE × *POST*	0.001	(0.31)	0.002	(0.54)
DE* × *POST* × *HIGHDP20	**0.064*** **	**(3.01)**		
DE* × *POST* × *HIGHDP33			**0.032** **	**(2.10)**
SIZE	0.005	(1.00)	0.003	(0.64)
LEV	0.071***	(2.68)	0.076***	(2.73)
LOSS	0.024***	(4.55)	0.026***	(5.00)
VOLA	−0.040	(−0.85)	−0.027	(−0.56)
NEWISSUE	−0.005	(−1.15)	−0.005	(−1.33)
AGE	0.000	(0.33)	0.000	(0.33)
RETURN	−0.002	(−0.76)	−0.006	(−1.45)
CLEV	0.067***	(3.86)	0.064***	(3.65)
LIQUIDITY	−0.034**	(−2.36)	−0.035**	(−2.33)
OCF	−0.063**	(−2.10)	−0.064**	(−2.15)
BIGN	0.048***	(3.21)	0.046***	(3.05)
BETA	−0.001	(−0.53)	−0.001	(−0.57)
REPORTLAG	0.000**	(2.49)	0.000**	(2.57)
HIGHDP20	0.012	(1.62)		
HIGHDP33			−0.003	(−0.40)
Firm effects	Yes		Yes	
Year effects	Yes		Yes	
Observations	8,974		8,974	
Adjusted R^2	0.04		0.04	

Notes. This table presents the effect of the *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions using alternative insolvency cutoffs. Model (1) reports results using quintile cutoff of the default probability (*DP*) distribution. Model (2) reports results using tertile cutoffs of default probability (*DP*) distribution. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

Table 10. Alternative Insolvency Measures

	(1)		(2)	
Variable	GC	*t*-stat	GC	*t*-stat
DE × *POST*	0.005	(1.07)	0.007	(1.60)
DE* × *POST* × *HIGHDP-ZS	**0.034** **	**(1.99)**		
DE* × *POST* × *HIGHDP-LEV			**0.023** **	**(2.32)**
SIZE	0.001	(0.18)	−0.003	(−0.46)
LEV	0.063**	(2.24)		
LOSS	0.020***	(3.65)	0.026***	(5.06)
VOLA	−0.043	(−0.85)	−0.012	(−0.25)
NEWISSUE	−0.005	(−1.32)	−0.006	(−1.43)
AGE	0.000	(0.09)	0.001	(0.66)
RETURN	−0.005	(−1.54)	−0.006	(−1.59)
CLEV	0.062***	(3.30)	0.094***	(4.17)
LIQUIDITY	−0.031*	(−1.89)	−0.048***	(−2.94)
OCF	−0.067**	(−2.35)	−0.064**	(−2.13)
BIGN	0.048***	(3.08)	0.046***	(3.02)
BETA	−0.000	(−0.10)	−0.001	(−0.74)
REPORTLAG	0.000**	(2.46)	0.000***	(2.60)
HIGHDP-ZS	0.014	(1.55)		
HIGHDP-LEV			0.017*	(1.77)
Firm effects	Yes		Yes	
Year effects	Yes		Yes	
Observations	8,708		8,974	
Adjusted R^2	0.04		0.04	

Notes. This table presents the effect of the *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions using *ZSCORE* (model (1)) and *LEV* (model (2)) as alternative measures for insolvency. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

differential exposure to economic changes across industries. Results are reported in Table 11. We find that the inclusion of *Year* × *State* and *Year* × *Industry* fixed effects does not alter our inferences. Overall, our results suggest that our findings are unlikely to be driven by variations of the impacts of other economic events across geographic regions and industries.

6.5. Other Robustness Tests

Prior research suggests that media attention may affect auditors' tendency to issue GC opinions to potential bankrupt firms (Mutchler et al. 1997). Thus, to the extent the *Credit Lyonnais* ruling increased press coverage of near-insolvent Delaware firms (but not near-insolvent non-Delaware firms) and the auditors had incentives to respond to the increased press coverage, the increase in GC opinions following the ruling could be driven by an increase in media attention. To rule out this possibility, we hand-collect media-coverage data on the number of news reports from Factiva and add media attention (*MEDIA*) as an additional control variable to Equation (1). As shown in Table 12, we find that the coefficient on *MEDIA* is insignificant,[9] and our main coefficient of interest remains statistically significant after controlling for *MEDIA*.

Aier et al. (2014) find an increase in accounting conservatism after the ruling.[10] To ensure that our findings of increase in GC opinions after the ruling are not confounded by increase in conservatism after the ruling, we examine the robustness of our results to controlling for conservatism. Following Aier et al. (2014), we measure conservatism (*CRANK*) as the average of decile ranks of three individual conservatism measures: negative nonoperating accruals, earnings skewness, and unrecorded reserves. The results are reported in Table 13. We find that our results remain unaffected.

We also use the propensity-score matching (PSM) method as an alternative approach to control for endogeneity. Results are reported in Tables 14 and 15. Table 14 shows results of first-stage probit regression and the covariates balance test,[11] which indicates that all covariates are balanced with no significant differences after matching. As shown in Table 15, our inferences remain unaltered using the propensity-score matched sample.

Table 11. Controlling for Variations Across Geographic Regions and Industries

	GC	
Variable	Coefficient	*t*-stat
DE × *POST*	0.003	(0.55)
DE* × *POST* × *HIGHDP	**0.059*****	**(2.85)**
SIZE	0.005	(0.88)
LEV	0.068**	(2.57)
LOSS	0.024***	(4.01)
VOLA	−0.036	(−0.71)
NEWISSUE	−0.005	(−1.41)
AGE	−0.000	(−0.00)
RETURN	−0.006	(−1.51)
CLEV	0.066***	(4.00)
HIGHDP	−0.001	(−0.08)
LIQUIDITY	−0.044***	(−2.91)
OCF	−0.060**	(−2.02)
BIGN	0.044***	(2.86)
BETA	−0.001	(−0.54)
REPORTLAG	0.000***	(2.84)
Firm effects	Yes	
Year × state effects	Yes	
Year × industry effects	Yes	
Observations	8,953	
Adjusted R^2	0.05	

Notes. This table presents robustness checks for the effect of *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions after controlling for variation across geographic regions and industries. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

6.6. Moderating Effect of Corporate Governance

Prior literature suggests that strong corporate governance mitigates management pressure on auditors to withhold GC opinions (e.g., Carcello and Neal 2000). To the extent that strong corporate governance substitutes for fiduciary duties as a mechanism to mitigate management pressure on auditors, firms with strong corporate governance are less likely to be affected by the ruling. In other words, the impact of the ruling on GC opinions may be weaker for firms with strong corporate governance. To test this possibility, we obtain data on outside director percentage from Compact Disclosure CD-ROM and construct a measure of director independence, *OUTDIR*, defined as the percentage of the board of directors that are not also officers of the firm.[12] We estimate Equation (1) after including *OUTDIR* and *DE* × *POST* × *HIGHDP* × *OUTDIR*. In untabulated results, we find that the coefficient on *DE* × *POST* × *HIGHDP* is significantly positive, and the coefficient on *DE* × *POST* × *HIGHDP* × *OUTDIR* is insignificantly negative. Although the negative coefficient is directionally consistent with the ruling having a weaker effect on firms with strong governance by outside directors, the evidence is generally inconclusive.

Table 12. Other Robustness Tests: Controlling for Media Attention

Variable	GC	*t*-stat
DE × *POST*	0.000	(0.00)
DE* × *POST* × *HIGHDP	**0.056*****	**(3.10)**
SIZE	0.006	(1.18)
LEV	0.072***	(2.73)
LOSS	0.025***	(4.57)
VOLA	−0.034	(−0.68)
NEWISSUE	−0.005	(−1.19)
AGE	0.001	(0.47)
RETURN	−0.004	(−1.26)
CLEV	0.066***	(3.75)
HIGHDP	0.001	(0.07)
LIQUIDITY	−0.034**	(−2.35)
OCF	−0.066**	(−2.20)
BIGN	0.048***	(3.15)
BETA	−0.001	(−0.51)
REPORTLAG	0.000***	(2.64)
MEDIA	−0.004	(−0.93)
Firm effects	Yes	
Year effects	Yes	
Observations	8,974	
Adjusted R^2	0.04	

Notes. Tables 12, 13, 14, and 15 present three robustness tests. This table presents the effect of *Credit Lyonnais* ruling on the likelihood of firms receiving GC opinions after controlling for media coverage. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

6.7. The Effect of the Ruling on the Likelihood of Dismissing Auditors Issuing GC Reports

As mentioned earlier, prior literature suggests that one important way directors and managers can influence auditor opinions is to dismiss the auditor who issues a GC report (e.g., Mutchler 1984 and Carcello and Neal 2003). If the ruling indeed reduces directors' and officers' incentives to pressure auditors to withhold their GC opinions, we expect that, after the ruling, auditors are less likely to be dismissed following the issuance of a GC report. To test this prediction, we conduct a difference-in-differences test using the following regression model:

$$Prob(DISMISS_{it} = 1) = \Lambda(\beta_0 + \beta_1 DE_{it} + \beta_2 POST_{it} + \beta_3 DE_{it} \times POST_{it} + \gamma Q_{it} + \varepsilon_{it}), \quad (2)$$

where $\Lambda[.]$ is the probit cumulative distribution function, and the subscripts i and t refer to firm and year, respectively. The dependent variable, *DISMISS*, is an indicator variable equal to one if the firm receiving a GC opinion in year t dismissed its auditor in year $t+1$, and zero otherwise. *DE* is an indicator variable for Delaware firms, and *POST* is an indicator variable equal to one (zero) for firm-years in the postruling (preruling) window. We expect the coefficient on the interaction, *DE* × *POST*, to be negative ($\beta_3 < 0$),

Table 13. Other Robustness Tests: Controlling for Reporting Conservatism

Variable	GC	*t*-stat
DE × *POST*	−0.012	(−1.11)
DE × *POST* × *CRANK*	0.003	(1.19)
DE* × *POST* × *HIGHDP	**0.057*** **	**(3.19)**
CRANK	0.001	(0.53)
SIZE	0.005	(1.08)
LEV	0.071***	(2.61)
LOSS	0.025***	(4.41)
VOLA	−0.036	(−0.73)
NEWISSUE	−0.005	(−1.16)
AGE	0.000	(0.25)
RETURN	−0.004	(−1.18)
CLEV	0.067***	(3.85)
HIGHDP	0.000	(0.02)
LIQUIDITY	−0.035**	(−2.44)
OCF	−0.066**	(−2.19)
BIGN	0.048***	(3.13)
BETA	−0.001	(−0.52)
REPORTLAG	0.000***	(2.65)
Firm effects	Yes	
Year effects	Yes	
Observations	8,953	
Adjusted R^2	0.04	

Notes. Tables 12, 14, and 15 and this table present three robustness tests. This table presents the effect of *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions after controlling for reporting conservatism (*CRANK*). Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

indicating that, relative to near-insolvent non-Delaware firms, near-insolvent Delaware firms are less likely to dismiss their auditors following a GC report after the ruling. For the *Q* vector of controls, we follow prior studies (e.g., Carcello and Neal 2003) and include a number of cross-sectional determinants of auditor dismissals, including firm size (*SIZE*), default probability based on the Merton (1974) model (*DP*), auditor tenure (*TENURE*), and an auditor's industry share, as in Hogan and Jeter (1999) (*INDSHARE*). We include industry fixed effect and cluster standard errors by the interaction of year and state of incorporation.[13]

To identify auditor dismissals following GC reports, we start with all near-insolvent firm-years with a GC report, and then read the SEC filing and press releases to identify whether the auditor was dismissed in the following year. Of the 158 near-insolvent firm-years with a GC report, we find that 11 were followed by an auditor dismissal in the following year. Table 16 reports a univariate difference-in-differences test. We compare *DISMISS* in the post-ruling period with those in the preruling period for treatment (near-insolvent Delaware) group and control (near-insolvent non-Delaware) group separately. We then calculate difference-in-differences of *DISMISS* by comparing the difference in the treatment group with

Table 14. Other Robustness Tests: Propensity-Score Matching (First-Stage Probit and Test of Covariates Balance)

	First-stage probit		Test of covariates balance		
Variable	GC	*t*-stat	Treatment	Control	*p*-value
SIZE	0.014	(0.51)	4.041	4.128	0.535
LEV	0.335**	(2.35)	0.845	0.815	0.399
LOSS	0.836***	(8.47)	0.903	0.903	1.000
VOLA	1.008**	(2.32)	0.197	0.195	0.872
NEWISSUE	−0.126*	(−1.66)	0.599	0.642	0.256
AGE	0.004	(1.19)	15.263	14.594	0.469
RETURN	−0.192**	(−2.42)	−0.284	−0.278	0.860
CLEV	0.458*	(1.88)	0.146	0.128	0.341
HIGHDP	0.304***	(3.21)	0.728	0.750	0.530
LIQUIDITY	−1.107***	(−3.21)	0.079	0.079	0.990
OCF	−0.687***	(−3.59)	−0.069	−0.076	0.754
BIGN	0.201	(1.62)	0.894	0.917	0.302
BETA	−0.023	(−1.27)	0.661	0.695	0.838
REPORTLAG	0.003***	(4.72)	93.198	95.770	0.636
Constant	−3.421***	(−6.00)			
Industry effects	Yes				
Observations	8,824				
Pseudo R^2	0.29				

Notes. Tables 12, 14, and 15 and this table present three robustness tests. This table and Table 15 present the effect of the *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions using propensity score matching as an alternative research design strategy. This table reports the first-stage probit regression and test of covariates balance. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

that in the control group. The Diff-in-diff variable is negative and significant (Diff-in-diff = −0.186 *p*-value = 0.025), supporting our conjecture.

Regression analyses of estimating Equation (2) are reported in Table 17. We find that our variable of interest, *DE* × *POST*, is negative and statistically significant (coefficient = −1.733, *t*-stat = −2.77), indicating

Table 15. Other Robustness Tests: Propensity-Score Matching (Second-Stage PSM Result)

Variable	GC	*t*-stat
DE × *POST*	−0.156	(−0.85)
DE* × *POST* × *HIGHDP	**0.433** **	**(2.52)**
HIGHDP	−0.056	(−0.81)
Firm effects	Yes	
Year effects	Yes	
Observations	868	
Adjusted R^2	0.02	

Notes. Tables 12, 13, and 14 and this table present three robustness tests. Table 14 and this table present the effect of *Credit Lyonnais* ruling on the likelihood of firms receiving auditors' GC opinions using propensity score matching as an alternative research design strategy. This table reports the second-stage results using the propensity-score-matched sample. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

Table 16. The Effect of the Ruling on the Likelihood of Dismissing Auditors Issuing GC Reports for Near-Insolvent Subsample: Univariate Analysis

	Delaware			Non-Delaware				
Variable	Pre	Post	*p*-value	Pre	Post	*p*-value	Diff-in-diff	*p*-value
Firm-years	36	54		39	29			
DISMISS	0.083	0.019	0.147	0.051	0.172	0.107	–0.186	0.025

Notes. This table and Table 17 present the effect of the ruling on the likelihood of dismissing auditors issuing GC reports for near-insolvent subsample. This table presents univariate analysis. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

Table 17. The Effect of the Ruling on the Likelihood of Dismissing Auditors Issuing GC Reports for Near-Insolvent Subsample: Regression Analysis

	DISMISS	
Variable	Coefficient	*t*-stat
DE	0.282	(0.64)
POST	0.607	(1.25)
DE × POST	**–1.733*** **	**(–2.77)**
SIZE	–0.001	(–0.01)
TENURE	0.009	(0.20)
DP	–0.749	(–1.02)
INDSHARE	2.389	(1.50)
Industry effects	Yes	
Observations	158	
Pseudo R^2	0.17	

Notes. Table 16 and this table present the effect of the ruling on the likelihood of dismissing auditors issuing GC reports for near-insolvent subsample. This table presents regression analysis. Standard errors are corrected for heteroscedasticity and are clustered by the interaction of year and state of incorporation. All variables are defined in the appendix.

*, **, and *** denote two-tailed significance level at 10%, 5%, and 1%, respectively.

a decrease in the likelihood of dismissing an auditor following the issuance of a GC report after the ruling for near-insolvent firms incorporated in Delaware relative to near-insolvent firms incorporated elsewhere.[14]

7. Conclusion

We explore the 1991 Delaware court ruling in the case of *Credit Lyonnais v. Pathe Communication* as an exogenous shock to managerial fiduciary duties to examine the causal effect of managerial fiduciary duties on the likelihood of firms receiving GC opinions. Prior to the ruling, directors and officers owed fiduciary duties to shareholders for solvent firms. The ruling expanded their fiduciary duties to creditors and other stakeholders for Delaware firms that are solvent, but in the "zone of insolvency." We find an increase in the likelihood of receiving GC reports from auditors after the ruling for near-insolvent Delaware firms relative to near-insolvent non-Delaware firms. In further tests, we find an increase in type I audit-opinion errors, but no change in audit risk after the ruling. Additionally, relative to near-insolvent non-Delaware firms, near-insolvent Delaware firms become less likely after the ruling to dismiss their auditors following the issuance of a GC report. Overall, our findings are consistent with managers with increased fiduciary duties toward creditors exerting less pressure on auditors and allowing them to reveal more GC opinions.

Our study contributes to the growing literature studying the effects of managerial fiduciary duties by providing new evidence on third-party consequences of managerial fiduciary duties. We also contribute to our understanding of auditors' GC opinions by examining the impact of an important, previously unidentified governance mechanism—managerial fiduciary duties. Our findings should be of interest to regulators and investors who are interested in understanding this important corporate-governance mechanism.

Acknowledgments

The authors appreciate helpful comments from an associate editor, Linda Bamber, Denny Beresford, Jenny Gaver, Ross Jennings, John (Xuefeng) Jiang, Sok-Hyon Kang, Miguel Minutti-Meza, Jaime Schmidt, two anonymous reviewers, and workshop participants at the University of Georgia, the 2016 AAA Annual meeting, the 2017 Auditing Section Midyear Meeting and the 2017 AAA FARS Midyear Meeting. This paper was previously presented under the title "Equity-Debt Agency Costs and Demand for Audit Quality." The authors are especially grateful to Ted Christensen for his invaluable help with this paper. The authors thank Robin Smith and Olivia Parker for collecting the media coverage data.

Appendix. Variable Definitions

Variable	Definition
AGE	The number of years of data for a firm since the coverage in Compustat.
AUCHANGE	An indicator variable that takes a value of one if the firm engages a new auditor, and zero otherwise.
BETA	A firm's beta estimated using a market model over the fiscal year
BIGN	An indicator variable that takes a value of one if a firm is audited by a Big N audit firm (Compustat item AU is between 1 and 8), and zero otherwise.
BUSY	An indicator variable that takes a value of one if the firm's fiscal year end is in December and January, and zero otherwise.
CLEV	Change in *LEV* during the year.
CRANK	A composite measure of conservatism, as in Aier et al. (2014). It is the simple average of decile ranks of three individual conservatism measures: negative nonoperating accruals, earnings skewness, and unrecorded reserves on the balance sheet.
DA	Discretionary accruals estimated from the cross-sectional version of the performance-adjusted modified Jones model.
DE	An indicator variable that takes a value of one if a firm is incorporated in Delaware, and zero otherwise.
DISMISS	An indicator variable that takes a value of one if the firm receiving a GC opinion in year I dismissed its auditor in year $t+1$, and zero otherwise.
DP	A firm's default probability based on the Merton (1974) model.
FAILURE	An indicator variable that takes a value of one if the firm either filed for bankruptcy or was delisted from the stock exchange (delisting code is 4 or 5) within one year of an audit, and zero otherwise.
FSALE	Sales from foreign operation to the firm's total sales.
GC	An indicator variable that takes a value of one if a firm receives a first-time going-concern audit opinion, and zero otherwise.
HIGHDP	An indicator variable that takes a value of one if a firm's *DP* is in the top quartile of the distribution, and zero otherwise.
HINDEX	The sum of the squared value of the ratio of operating segment sales to total sales.
INDSHARE	An auditor's industry share as in Hogan and Jeter (1999), which is the sum of the square root of assets of all firms an auditor audited in a given two-digit Standard Industrial Classification (SIC) code divided by the sum of the square root of assets across all Compustat firms in the same two-digit SIC code.
LEV	Total liabilities over total assets.
LIQUIDITY	The sum of a firm's cash and investment securities (long and short term) over total assets.
LOSS	An indicator variable that takes a value of one if a firm's income before extraordinary items is negative, and zero otherwise.
MEDIA	The natural logarithm of one plus the number of news reports of a firm over a fiscal year from Factiva.
NEWISSUE	An indicator variable that takes a value of one if a firm has a new issuance of equity or debt in the subsequent year (i.e., nonzero Compustat item DLTIS or the amount of Compustat item SSTK is over 5% of the firm's market value of equity), and zero otherwise.
OCF	Operating cash flows over total assets.
OPINION	An indicator variable that takes a value of one if the auditor issues an unqualified opinion, and zero otherwise.
POST	An indicator variable that takes a value of one for the postruling period from 1992–1994, and zero for the preruling period from 1989–1991.
REPORTLAG	The number of days between the fiscal year end date and the earnings announcement date.
RETURN	A firm's stock return over the fiscal year.
ROA	Income before extraordinary items over total assets.
SIZE	The natural logarithm of total assets.
TENURE	The number of consecutive years that a client has retained an auditor.
VOLA	Standard deviation of a firm's monthly stock returns over the year.
ZSCORE	Altman's Z-score (Altman 1968), calculated as 1.2 (working capital/total assets) + 1.4 (retained earnings/total assets) + 3.3 (earnings before interest and taxes/total assets) + 0.6 (market value of equity/total liabilities) + (sales/total assets).

Endnotes

[1] In general, directors are indemnified by companies for litigation expenses and can take out insurance. But, as the company approaches insolvency, companies may not be able to indemnify. Also, insurance companies may not pay for dereliction of fiduciary duties (Becker and Strömberg 2012).

[2] The ruling on the 1991 Credit Lyonnais case was partially reversed by two recent Delaware cases (*Production Resources v. NCT* in 2004 and *North American Catholic Education Programming Foundation, Inc., v. Gheewalla* in 2007). The ruling of both cases held that creditors cannot sue directors and managers directly for breach of fiduciary duties, except for those in contractual arrangements, potentially weakening the relevance of managerial fiduciary duties to creditors created by the Credit Lyonnais case. However, these later cases received much less publicity compared with Credit Lyonnais, and, empirically, Becker and Strömberg (2012) find no change in their results around these later cases.

[3] Identifying near-insolvent firms also provides a more powerful test of auditors' GC opinions because the GC opinion decision is most salient among these firms (e.g., Hopwood et al. 1994, Mutchler et al. 1997, Reynolds and Francis 2000, and DeFond et al. 2002).

[4] In untabulated robustness checks, we find that our results are also robust to clustering standard errors by firm or by both firm and the interaction between year and state. Our results are also robust to controlling for a measure of director independence, defined as the percentage of the board of directors that are not also officers of the firm.

[5] We use fiscal year instead of calendar year to define our sample period. So, the post period will not start until June 1992. Also, auditors' GC decisions will not be available until a few more months later after the fiscal year end. Therefore, our research design is more conservative and works against finding results.

[6] We clarify three issues here. First, our search includes 1988 because we examine first-time GC, which requires a firm's audit opinion one

year prior to the current year to be observable. Second, Statement of Auditing Standards No. 59 (effective January 1, 1989) requires auditors' GC opinions to use specific wording that includes the terms "substantial doubt" and "going concern." Searching only "going concern" will allow us to cast a wider net to capture all potential GC opinions. Finally, we use the following search term in LexisNexis: (going concern) and (Annual Report or 10k or 10-k or 10ksb or 10-ksb or 10k405 or 10-k405) and (DOCUMENT-DATE AFT(XXXX) AND DOCUMENT-DATE BEF(YYYY)).

[7] We do not include *DE* or *POST*, as they are absorbed by the firm and year fixed effects, respectively. Moreover, because the coefficient on *HIGHDP* is almost zero, including the interaction with *DE* or *POST* seems unnecessary and may cause multicollinearity problems. However, in an untabulated robustness test, we add *DE* × *HIGHDP* and *POST* × *HIGHDP* to the regression model. We find that none of the two-way interactions are significant, and our variable of interest remains highly significant.

[8] Given the well-known incidental parameter problem for a probit model with firm fixed effects (e.g., Greene 2012), we replace the firm fixed effects with industry effects and run a modified Equation (1) as follows: $\text{Prob}(GC = 1) = \beta_0 + \beta_1 DE + \beta_2 DE \times POST + \beta_3 DE \times HIGHDP + \beta_4 POST \times HIGHDP + \beta_5 DE \times POST \times HIGHDP + \text{Controls} + \alpha_j + \lambda_t + \varepsilon$.

[9] Note that our test differs from Mutchler et al. (1997), who consider news content and examine news in four categories (i.e., extreme positive, mild positive, mild negative, and extreme negative) and find a positive relation between GC opinions and extreme negative news. In comparison, our *MEDIA* variable captures the overall media coverage and is measured by an overall count of all news reports. Thus, one likely reason for our finding of no significant relation between *MEDIA* and GC opinions is that the positive relation for extreme negative news is offset by less negative or no effect of other types of news (e.g., positive news).

[10] Our finding of more GC opinions after the ruling is not inconsistent with the findings of Aier et al. (2014), as both conservatism and GC opinions can be mechanisms to protect debtholders' interests in distressed firms. Thus, after the ruling increased managerial fiduciary duties toward debtholders, managers became more creditor-friendly and, thus, more likely to report more conservatively and/or put less pressure on auditors to withhold their GC opinions. In addition, prior research finds that after the ruling, managers also increased investment and equity issuance and decreased risks (Becker and Strömberg 2012)—other actions that managers took to protect creditors. Overall, these prior findings and our results combine to paint a consistent picture that the ruling led managers to enhance mechanisms that protect creditors.

[11] More specifically, we match each GC firm with non-GC firms based on the three nearest neighbors with equal weight, replacement, and a caliper of 0.2.

[12] Carcello and Neal (2000) examine the association between audit-committee characteristics and GC opinions. To our knowledge, data on audit committee characteristics is not available in machine-readable format for our sample period. Further, establishing an audit committee was not a universal requirement for all firms during our sample period. Assuming that the whole board of directors had the responsibilities of the audit committee, *OUTDIR* can capture some of the specific effect of an audit committee, plus a broader governance effect by the whole board.

[13] Unlike Equation (1), we do not include firm fixed effects because of the small sample size. Many firms do not experience any auditor changes throughout the sample period. Controlling for firm fixed effects will drop these observations and further shrink the sample size.

[14] Despite significant results, we caution readers that sample size for this test is small and that this result is limited to only near-insolvent firms that have received a GC report.

References

Aier J, Chen L, Pevzner M (2014) Debtholders' demand for conservatism: Evidence from changes in directors' fiduciary duties. *J. Accounting Res.* 52(5):993–1027.

Altman E (1968) Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *J. Finance* 23(4):589–609.

Amin K, Krishnan J, Yang J (2014) Going concern opinion and cost of equity. *Auditing* 33(4):1–39.

Becker B, Strömberg P (2012) Fiduciary duties and equity-debtholder conflicts. *Rev. Financial Stud.* 25(6):1931–1969.

Bertrand M, Mullainathan S (2003) Enjoying the quiet life? Corporate governance and managerial preferences. *J. Polit. Econom.* 111: 1043–1075.

Bens D, Huang S, Tan L, Wongsunwai W (2020) Contracting and reporting conservatism around a change in fiduciary duties. *Contemp. Accounting Res.* 37(4):2472–2500.

Bhagat S, Black B (2002) The non-correlation between board independence and long-term firm performance. *J. Corporate Law* 27:231–273.

Carcello J, Neal T (2000) Audit committee composition and auditor reporting. *Accounting Rev.* 75(October):453–467.

Carcello J, Neal T (2003) Audit committee characteristics and auditor dismissals following "new" going-concern reports. *Accounting Rev.* 78(January):95–117.

Carson E, Fargher N, Geiger M, Lennox C, Raghunandan K, Willekens M (2013) Audit reporting for going-concern uncertainty: A research synthesis. *Auditing* 32:353–384.

Chen K, Church B (1992) Default on debt obligations and the issuance of going concern opinions. *Auditing* 11:3–49.

Chen C, Martin X, Wang X (2013) Insider trading, litigation concerns, and auditor going-concern opinions. *Accounting Rev.* 88(2): 365–393.

Chen P, He S, Ma Z, Stice D (2016) The information role of audit opinions in debt contracting. *J. Accounting Econom.* 61(1): 121–144.

DeFond ML (1992) The association between changes in client firm agency costs and auditor switching. *Auditing* 11:16–31.

DeFond M, Zhang J (2014) A review of archival auditing research. *J. Accounting Econom.* 58(2-3):275–326.

DeFond M, Raghunandan K, Subramanyam K (2002) Do non-audit service fees impair auditor independence? Evidence from going-concern audit opinions. *J. Accounting Res.* 40(4):1247–1274.

Dionne A (2007) Living on the edge: Fiduciary duties, business judgment and expensive uncertainty in the zone of insolvency. *Stanford J. Law Bus. Finance* 13:188–224.

Dopuch N, Holthausen R, Leftwich R (1987) Predicting audit qualifications with financial and market variables. *Accounting Rev.* 62(3):431–454.

Francis J (2011) A framework for understanding and researching audit quality. *Auditing* 30(2):125–152.

Francis JR, Wilson ER (1988) Auditor changes: A joint test of theories relating to agency costs and auditor differentiation. *Accounting Rev.* 63(4):663–682.

Ghosh A, Tang C (2015) Assessing financial reporting quality of family firms: The auditors' perspective. *J. Accounting Econom.* 60:95–116.

Greene W (2012) *Econometric Analysis*, 7th ed. (Pearson, New York).

Gutiérrez M (2003) An economic analysis of corporate directors' fiduciary duties. *RAND J. Econom.* 34(3):516–535.

Hogan C, Jeter D (1999) Industry specialization by auditors. *Auditing* 18(1):1–17.

Holder-Webb L, Wilkins M (2000) The incremental information content of SAS No. 59 going-concern opinions. *J. Accounting Res.* 38(1):209–219.

Hopwood W, McKeown J, Mutchler J (1994) Reexamination of auditor vs. model accuracy within the context of the going-concern opinion decision. *Contemp. Accounting Res.* 11(2):295–310.

Hughes RL, McGee M (1995) Fiduciary duties of directors of insolvent corporations. Quittenden AM, ed. *Annual Current Developments in Bankruptcy and Reorganization* (Practicing Law Institute, New York), 6–10.

Lennox C (2000) Do companies successfully engage in opinion shopping? The UK experience. *J. Accounting Econom.* 29:321–337.

Menon K, Williams D (2010) Investor reaction to going concern audit reports. *Accounting Rev.* 85(6):2075–2105.

Merton RC (1974) On the pricing of corporate debt: The risk structure of interest rates. *J. Finance* 29(2):449–470.

Mutchler J (1984) Auditors' perceptions of the going-concern opinion decision. *Auditing* 3(2):17–30.

Mutchler J, Hopwood W, McKeown J (1997) The influence of contrary information and mitigating factors in audit opinion decisions on bankrupt companies. *J. Accounting Res.* 35:295–310.

Myers L, Schmidt J, Wilkins M (2014) An investigation of recent changes in going concern reporting decisions among Big N and Non-Big N auditors. *Rev. Quant. Finance Accounting* 43(1): 155–172.

Reynolds K, Francis J (2000) Does size matter? The influence of large clients on office-level auditor reporting decisions. *J. Accounting Econom.* 30(3):375–400.

Stanley J, Zoort F, Taylor G (2009) The association between insider trading surrounding going concern audit opinions and future bankruptcy. *Managerial Auditing J.* 24(3):290–312.

Vassalou M, Xing Y (2004) Default risk in equity returns. *J. Finance* 59(2):831–868.

Willenborg M, McKeown J (2000) Going-concern initial public offerings. *J. Accounting Econom.* 30(3):279–313.